

MATH203 – APPLIED LINEAR ALGEBRA

FINAL EXAM REVIEW GUIDE (2026 SPRING)

Scope of this guide. This guide covers the *post-midterm* material. The earlier material is still examinable; for that, use the *Midterm Review Guide*.

EXAM FORMAT

Category	Points	Percentage
Counterexamples	15	15%
Computation	65	65%
Proof	20	20%
Total	100	100%

- **Counterexamples:** just list the matrices requested. You do *not* need to verify the claimed properties — the graders will. Small (2×2) examples are preferred.
- **Computation:** present the answer *with* its derivation. Every final matrix must be written entry-wise; an answer left only as a product or sum of matrices receives no credit (unless a factorization such as $A = PDP^{-1}$, $A = U\Sigma V^T$, or $A = MM^T$ is what was asked for).
- **Proof:** all logic must be complete and closed. You may use only basic algebra and the properties or theorems explicitly stated in the problem. Basic algebra includes identities such as

$$A(B + C) = AB + AC, \quad (AB)^T = B^T A^T, \quad (AB)^{-1} = B^{-1} A^{-1},$$

and direct substitutions such as $AI = A$, $A0 = 0$, or $P^2 = P$ when P is given as a projection.

- Orthonormal bases (for SVD, for $\text{Col}(P)$) are built by *diagonal cross-filling*, not Gram–Schmidt, which has not been taught in this course.

SCOPE: POST-MIDTERM MATERIAL

Least Squares and Linear Regression.

- $W = \text{Col}(B)$, $P_W = B(B^T B)^{-1} B^T$
- $\hat{\mathbf{x}} = (B^T B)^{-1} B^T \mathbf{b}$, $B\hat{\mathbf{x}} = P_W \mathbf{b}$
- $\mathbf{b} - P_W \mathbf{b} \perp W$, $\min_{\mathbf{x}} \|\mathbf{b} - B\mathbf{x}\|^2 = \|\mathbf{b} - P_W \mathbf{b}\|^2$
- Compute $\hat{\mathbf{x}} = (B^T B)^{-1} B^T \mathbf{b}$ by simultaneous row reduction on $(B^T B, B^T \mathbf{b})$

Determinants.

- Cross-filling determinant: $\det(A) = a_1 \cdots a_n \det(S)$, where a_i are cross-centers and S is the pivot-position switching matrix
- $n \times n$ switching matrix, m zigzag loops $\implies \det(S) = (-1)^{n+m}$
- $\det(XY) = \det(X) \det(Y)$, $\det(A^T) = \det(A)$, $\det(A^{-1}) = 1/\det(A)$
- $\det \begin{pmatrix} X & * \\ 0 & Y \end{pmatrix} = \det(X) \det(Y)$
- Adjugate identity: $AA^* = \det(A)I$

Spectral Decomposition, Eigenvalues, and Polynomial Division.

- Eigenvalues are the roots of $\det(tI - A)$
- Cayley–Hamilton: $p_A(t) = \det(tI - A)$, then $p_A(A) = 0$.
- Polynomial division by any annihilating polynomial $f(t)$:

$$g(t) = Q(t)f(t) + R(t), \quad f(A) = 0 \implies g(A) = R(A).$$

- A is diagonalizable $\iff q(A) = 0$ where $q(t)$ is the square-free part (radical) of $\det(tI - A)$ after splitting into linear factors; Distinct roots in $p_A(t) \implies q(t) = p_A(t) \implies$ diagonalizable by Cayley–Hamilton

- Spectral decomposition: If A diagonalizable, then $A = \sum_{\lambda} \lambda P_{\lambda}$,

$$g(A) = \sum_{\lambda} g(\lambda) P_{\lambda}, \quad P_{\lambda} = L_{\lambda}(A) = \prod_{\substack{\mu \neq \lambda \\ \mu \text{ eigenvalue}}} \frac{A - \mu I}{\lambda - \mu}$$

- Lagrange interpolation polynomials at $\lambda_1, \dots, \lambda_k$:

$$L_k(t) = \prod_{j \neq k} \frac{t - \lambda_j}{\lambda_k - \lambda_j}, \quad L_k(\lambda_j) = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

- Eigenprojections for diagonalizable matrices are compatible: $P_k^2 = P_k$, $P_j P_k = 0$ ($j \neq k$), $\sum_k P_k = I$
- $A^m = \sum_k \lambda_k^m P_k$, $e^A = \sum_k e^{\lambda_k} P_k$, $e^{At} = \sum_k e^{\lambda_k t} P_k$
- $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(t) = e^{At} \mathbf{x}(0)$
- Eigenspace $E_{\lambda} = \text{Col}(P_{\lambda})$; cross-filling P_{λ} gives right/left eigenbasis
- Diagonalization notation: $A = PDP^{-1}$; columns of P and rows of P^{-1} come from cross-filling the spectral projections
- Orthogonal diagonalization of a symmetric matrix: $A = QDQ^T$ with $Q^T Q = I$; the orthonormal basis in Q comes from diagonal cross-filling of the spectral decomposition
- Real matrices: non-real eigenvalues occur in conjugate pairs $a \pm bi$, non-real eigenprojections occur in conjugate pairs $P_{a \pm bi} = X \pm iY$

Normal Matrices $AA^H = A^H A$.

- Hermitian $A = A^H$, normal $AA^H = A^H A$, skew-Hermitian $X^H = -X$, unitary $U^H U = I$
- Normality criterion: A is normal iff it is diagonalizable and distinct eigenspaces are Hermitian-orthogonal
- Proof reductions:
 - $M^H M = 0 \Rightarrow M = 0$
 - $N^m = 0$ and $N^H N = N N^H \Rightarrow N = 0$
 - $P^2 = P$ and $P^H P = P P^H \Rightarrow P = P^H$
- For normal matrices, spectral projections are Hermitian-orthogonal projections (compatible projections: $P_i P_j = 0$ for $i \neq j$ and $\sum_i P_i = I$)
- Unitary diagonalization: diagonal cross-fill the spectral projection of normal matrices to collect an orthonormal eigenbasis to write $A = \Omega D \Omega^H$
- Special normal matrices are eigenvalue-location restrictions: Hermitian $\Leftrightarrow$ real eigenvalues, skew-Hermitian $\Leftrightarrow$ imaginary eigenvalues, unitary $\Leftrightarrow$ eigenvalues on the unit circle, positive definite $\Leftrightarrow$ Hermitian with all eigenvalues > 0 , positive semidefinite $\Leftrightarrow$ Hermitian with all eigenvalues ≥ 0

Positive (Semi)Definite Matrices.

- Positive definite: $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq 0$; positive semidefinite (PSD): ≥ 0
- Factorization from positive pivots: $A = M M^T$, with columns $\mathbf{c}_k / \sqrt{a_k}$, produced by the diagonal cross-filling technique
- Eigenvalue estimates: $\lambda_{\min} \|\mathbf{x}\|^2 \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_{\max} \|\mathbf{x}\|^2$
- Congruence: if A is PD and C has full column rank, then $C^T A C$ is PD; if A is PSD, then $C^T A C$ is PSD for any C

Diagonal Cross-Filling Positivity Test for an $n \times n$ Matrix.

Diagonal cross-filling output	Conclusion
No difficulties; n pivots, all > 0	Positive definite
No difficulties; all pivots are ≥ 0	Positive semidefinite
No difficulties; n pivots, all < 0	Negative definite
No difficulties; all pivots are ≤ 0	Negative semidefinite
No difficulties; pivots contain both positive and negative signs	Indefinite
Difficulty: zero diagonal pivot but nonzero row/column remains	Indefinite

Singular Value Decomposition.

- SVD $A = U\Sigma V^T$ with $U^T U = I$, $V^T V = I$, Σ “diagonal” with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$
- Right singular vectors: orthonormal eigenvectors of $A^T A$ from spectral decomposition / cross-filling; singular values $\sigma_i = \sqrt{\lambda_i(A^T A)}$
- Left singular vectors: $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$; complete U to an orthonormal basis by cross-filling $I - \sum_i \mathbf{u}_i \mathbf{u}_i^T$
- For a Hermitian (or symmetric) matrix, the singular values are the absolute values $|\lambda_i|$ of the eigenvalues

Key Formulas and Identities.

- Least-squares projection: $P_W = B(B^T B)^{-1} B^T$ and $\hat{\mathbf{x}} = (B^T B)^{-1} B^T \mathbf{b}$
- Cross-filling determinant: $\det(A) = a_1 \cdots a_n \det(S)$; for switching matrices, $\det(S) = (-1)^{n+m}$ from zigzag loops
- Polynomial division by an annihilating polynomial: if $g(t) = Q(t)f(t) + R(t)$ and $f(A) = 0$, then $g(A) = R(A)$
- Lagrange / spectral projection: $P_k = \prod_{j \neq k} \frac{A - \lambda_j I}{\lambda_k - \lambda_j}$, and $f(A) = \sum_k f(\lambda_k) P_k$
- Value table principle: same value table $\Rightarrow$ same polynomial in A ; products multiply value tables entry-wise
- Diagonalizability: if $q(t)$ is the square-free part (radical) of $\det(tI - A)$, then A is diagonalizable iff $q(A) = 0$ after splitting
- Matrix exponential: $e^A = \sum_k e^{\lambda_k} P_k$. **Warning:** $e^A e^B = e^{A+B}$ *only* when $AB = BA$
- Normal matrices: $AA^H = A^H A$; normal $\Leftrightarrow$ diagonalizable with Hermitian-orthogonal eigenspaces; special normal classes are eigenvalue-location tests
- Positive definite estimates: $\lambda_{\min} \|x\|^2 \leq x^T A x \leq \lambda_{\max} \|x\|^2$ for real symmetric A
- SVD recipe: diagonalize $A^T A = V D V^T$, set $\sigma_i = \sqrt{D_{ii}}$, $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$
- Positivity lemma (proof workhorse): $M^H M = 0 \iff M = 0$
- Determinant identity: $\det(I_n - AB) = \det(I_m - BA)$
- Rank (Sylvester) identity: $n - \text{rank}(I_n - AB) = m - \text{rank}(I_m - BA)$
- Both follow from the block matrix $S = \begin{pmatrix} I_n & A \\ B & I_m \end{pmatrix}$ factored two ways (block elimination on each side)

Computation Checklist.

- (1) Least-squares / orthogonal-projection data fitting using $P_W = B(B^T B)^{-1} B^T$ and simultaneous row operations for coefficients
- (2) Computing determinants by general cross-filling: record cross-centers, build the switching matrix, and use zigzag for its sign
- (3) Using Cayley–Hamilton and polynomial division to replace $g(A)$ by a low-degree remainder
- (4) Building spectral projections by value tables / Lagrange polynomials, then using them to compute A^m , e^A , e^{At}
- (5) Testing diagonalizability by plugging A into the square-free part (radical) of the characteristic polynomial
- (6) Writing diagonalization $A = PDP^{-1}$ from the spectral projections

- (7) Classifying normal / Hermitian / skew-Hermitian / unitary matrices by spectral geometry and eigenvalue location
- (8) Testing positive (semi)definiteness by diagonal cross-filling, zero-row behavior, and producing the $A = MM^T$ factorization
- (9) SVD from spectral decomposition of $A^T A$ and cross-filling completion of U
- (10) Determinants via cross-filling, Laplace expansion, and the block-triangular rule
- (11) Proof formulas: $M^H M = 0 \Rightarrow M = 0$, $AA^H = A^H A$, and the block-matrix det/rank identities

Common Mistakes to Avoid.

- Writing $e^A e^B = e^{A+B}$ when A and B do not commute
- Forgetting to sort eigenvalues / singular values in descending order
- Confusing singular values with eigenvalues (they coincide only up to absolute value, and only for Hermitian matrices)
- Assuming a real matrix that is diagonalizable over $\mathbb{R}$ with real eigenvalues must be symmetric (false)
- Forgetting that, for a real matrix, non-real eigenvalues come in conjugate pairs $a \pm bi$
- Reaching for Gram–Schmidt (not taught) instead of cross-filling to get orthonormal bases
- Declaring a matrix PSD when a diagonal pivot is 0 while an off-diagonal entry in its row is nonzero
- Leaving a final answer as a product/sum of matrices instead of writing it entry-wise